

Customer Churn Prediction Using Machine Learning

Codveda Technology – Data Analytics Internship

Submitted By

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Objective

The objective of this project is to develop and evaluate machine learning models that can predict whether a customer is likely to churn based on telecom customer data.

Dataset

Dataset Name: Customer Churn Dataset

The dataset contains customer information such as account length, international plan, voice mail plan, number of calls, call charges, and customer churn status.

Tools Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
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Machine Learning Algorithms

1. Logistic Regression
 2. Decision Tree Classifier
 3. Random Forest Classifier
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Methodology

Step 1 – Data Loading

Training and testing datasets were loaded into Pandas DataFrames.

Step 2 – Data Preprocessing

- Missing values were checked.
- Unnecessary columns were removed.
- Categorical values were encoded into numerical values.

Step 3 – Feature Selection

The **Churn** column was selected as the target variable, while the remaining columns were used as input features.

Step 4 – Model Training

Three classification models were trained:

- Logistic Regression
- Decision Tree
- Random Forest

Step 5 – Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Step 6 – Hyperparameter Tuning

GridSearchCV was applied to optimize the Random Forest model.

Results

Model	Accuracy
Logistic Regression	85.76%
Decision Tree	91.30%
Random Forest	95.65%

Random Forest achieved the highest accuracy and was selected as the best-performing model.

Conclusion

Customer churn prediction was successfully performed using multiple machine learning algorithms. Among all the models tested, the Random Forest Classifier provided the best performance with an accuracy of **95.65%**, making it the most suitable model for predicting customer churn in this dataset.